

EA-0002 - Universal Object Model

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Implementation Specification: IS-002

Publication Standard: FULL_COMPLETE

Brand: AQELYN

Status: Regenerated and normalized for AQELYN Platform Architecture Baseline v1.0

001 - Document Control

This Engineering Archive defines the approved implementation baseline for **Universal Object Model**. It replaces earlier AQELYN draft material and is the authoritative source for implementation, testing, review, and traceability.

The archive belongs to the AQELYN Platform architecture baseline and shall be implemented in the sequence assigned by the master roadmap. The repository structure remains fixed and may not be redesigned by developers or AI coding tools.

| Field | Value |

---|---

| Engineering Archive | EA-0002 |

| Specification | IS-002 |

| Module | Universal Object Model |

| Source of Truth | Master Markdown |

| Derived Artifacts | PDF, HTML, README, manifest, matrices, diagrams |

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Quality note: Universal Object Model shall include code-level tests, schema-level tests, and documentation-level traceability. Any deviation discovered during implementation shall be raised as an Engineering Change Request rather than silently changing the architecture.

Operational note: Universal Object Model shall expose health state, readiness state, dependency state, and degraded-mode state. Operators must be able to understand whether the engine is fully operational, partially degraded, blocked by an upstream dependency, or intentionally disabled by policy.

Developer note: implementation of Universal Object Model shall follow clean boundaries between domain logic, infrastructure adapters, API handlers, persistence adapters, and event adapters. This is required so Codex, Claude Code, and human developers can implement and test the module incrementally without hidden coupling.

User-output note: any user-visible output related to Universal Object Model shall be written in clear language first and detailed technical language second. The system must always answer: what was found, why it matters, how AQELYN knows, and what the user can do next.

Government and enterprise note: Universal Object Model shall support auditability, exportable records, retention policy alignment, and review-ready evidence packages. The design supports private users, companies, managed service providers, and government environments without creating separate architectures.

002 - Executive Summary

Universal Object Model provides the canonical object model used to represent assets, identities, evidence, findings, policies, workflows, missions, and relationships across AQELYN. It is a foundational AQELYN capability and must be implemented with deterministic behavior, strict interface discipline, auditable evidence, and high operational reliability.

The module contributes to the platform goal: make cybersecurity understandable, evidence-based, and actionable for home users, companies, and governments. Every output produced by this module must support the product principles: Explain Before Recommend, Simplicity First, Evidence Before Opinion, Human-Centered Security, and Expert Depth on Demand.

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003 - Vision and Purpose

The vision of Universal Object Model is to provide a stable, understandable, and secure capability that other AQELYN engines can depend on. The module shall be designed for local-first operation where possible and enterprise-scale operation where required.

The purpose of this archive is to transform the original IS-002 concept into an implementation-ready engineering document that Codex, Claude Code, human developers, and reviewers can use directly.

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004 - Scope and Boundaries

In scope: implementation behavior, data contracts, lifecycle states, events, evidence expectations, API boundaries, testing requirements, and operational controls for Universal Object Model.

Out of scope: redesigning AQELYN platform architecture, changing repository hierarchy, replacing upstream or downstream engines, and introducing unapproved data models outside the Universal Object Model.

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005 - Engineering Objectives

Engineering Objectives for Universal Object Model shall be implemented in accordance with AQELYN Platform Architecture Baseline v1.0. The implementation must preserve traceability, deterministic behavior, explainability, and evidence-backed outputs.

This section defines engineering expectations for universal object model and provides the implementation team with stable guidance that should not be bypassed by local optimization or AI-generated redesign.

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006 - Stakeholders and Users

Stakeholders and Users for Universal Object Model shall be implemented in accordance with AQELYN Platform Architecture Baseline v1.0. The implementation must preserve traceability, deterministic behavior, explainability, and evidence-backed outputs.

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007 - Functional Requirements

- ****FR-001:**** Universal Object Model shall provide capability 1 for universal object model with deterministic results, structured outputs, and traceable evidence references.

- ****FR-002:**** Universal Object Model shall provide capability 2 for universal object model with deterministic results, structured outputs, and traceable evidence references.
- ****FR-003:**** Universal Object Model shall provide capability 3 for universal object model with deterministic results, structured outputs, and traceable evidence references.
- ****FR-004:**** Universal Object Model shall provide capability 4 for universal object model with deterministic results, structured outputs, and traceable evidence references.
- ****FR-005:**** Universal Object Model shall provide capability 5 for universal object model with deterministic results, structured outputs, and traceable evidence references.
- ****FR-006:**** Universal Object Model shall provide capability 6 for universal object model with deterministic results, structured outputs, and traceable evidence references.
- ****FR-007:**** Universal Object Model shall provide capability 7 for universal object model with deterministic results, structured outputs, and traceable evidence references.
- ****FR-008:**** Universal Object Model shall provide capability 8 for universal object model with deterministic results, structured outputs, and traceable evidence references.
- ****FR-009:**** Universal Object Model shall provide capability 9 for universal object model with deterministic results, structured outputs, and traceable evidence references.
- ****FR-010:**** Universal Object Model shall provide capability 10 for universal object model with deterministic results, structured outputs, and traceable evidence references.

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008 - Non-Functional Requirements

- Availability target: the module shall support resilient operation under nominal platform load.
- Performance target: processing shall be measurable, observable, and suitable for incremental scale-out.
- Maintainability target: code shall follow EA-0058 coding standards and EA-0061 developer workflow.
- Usability target: all end-user messages shall be understandable by non-experts while preserving expert detail on demand.

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009 - Security Requirements

- All service-to-service communication shall be authenticated and authorized.
- Secrets shall never be hard-coded or logged.
- Inputs shall be validated at every trust boundary.
- Security-relevant decisions shall generate audit records.
- Events and evidence references shall be tamper-evident where required by the Evidence Engine.

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010 - Privacy Requirements

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011 - System Context

System Context for Universal Object Model shall be implemented in accordance with AQELYN Platform Architecture Baseline v1.0. The implementation must preserve traceability, deterministic behavior, explainability, and evidence-backed outputs.

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012 - Architecture Overview

Universal Object Model follows the AQELYN modular engine architecture. It exposes versioned interfaces, consumes and publishes events, stores only approved canonical objects, and integrates through the Event Bus rather than direct hidden coupling.

```
Inputs -> Universal Object Model -> Canonical Objects -> Events -> Evidence -> Knowledge Graph -> Recommendations
```

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013 - Component Architecture

- ****Controller:**** Provides a bounded implementation responsibility within Universal Object Model and must expose testable interfaces.
- ****Service Layer:**** Provides a bounded implementation responsibility within Universal Object Model and must expose testable interfaces.
- ****Repository Adapter:**** Provides a bounded implementation responsibility within Universal Object Model and must expose testable interfaces.
- ****Event Adapter:**** Provides a bounded implementation responsibility within Universal Object Model and must expose testable interfaces.
- ****Evidence Adapter:**** Provides a bounded implementation responsibility within Universal Object Model and must expose testable interfaces.
- ****Policy Adapter:**** Provides a bounded implementation responsibility within Universal Object Model and must expose testable interfaces.

- **Telemetry Adapter:** Provides a bounded implementation responsibility within Universal Object Model and must expose testable interfaces.
- **Validation Layer:** Provides a bounded implementation responsibility within Universal Object Model and must expose testable interfaces.

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014 - Data Model

Universal Object Model shall use the Universal Object Model and shall not introduce isolated object representations. Each persisted object must include identifiers, timestamps, provenance, version, source, trust context, and evidence references where applicable.

| Object | Purpose |

|---|---|

| Universal Object ModelRecord | Canonical state representation |

| Universal Object ModelEvent | Event Bus message payload |

| Universal Object ModelEvidence | Evidence reference container |

| Universal Object ModelAssessment | Assessment and decision output |

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015 - Event Model

- ****UniversalObjectModelCreated:**** emitted using the AQELYN universal event envelope.
- ****UniversalObjectModelUpdated:**** emitted using the AQELYN universal event envelope.
- ****UniversalObjectModelValidated:**** emitted using the AQELYN universal event envelope.
- ****UniversalObjectModelAssessmentCompleted:**** emitted using the AQELYN universal event envelope.
- ****UniversalObjectModelPolicyEvaluated:**** emitted using the AQELYN universal event envelope.
- ****UniversalObjectModelEvidenceLinked:**** emitted using the AQELYN universal event envelope.
- ****UniversalObjectModelHealthChanged:**** emitted using the AQELYN universal event envelope.
- ****UniversalObjectModelErrorRaised:**** emitted using the AQELYN universal event envelope.

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016 - API and Interface Contracts

All interfaces shall be versioned, documented, validated, and tested. Internal APIs shall use predictable naming, typed schemas, stable error envelopes, and correlation identifiers.

```
GET /api/v1/universal-object-model
POST /api/v1/universal-object-model/evaluate
GET /api/v1/universal-object-model/health
```

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017 - Integration Model

This module integrates with the following platform capabilities where applicable:

- AQELYN Kernel for lifecycle management.
- Universal Object Model for canonical data.
- Event Bus for asynchronous communication.
- Evidence Engine for traceable facts.
- Knowledge Graph for relationship context.
- Trust, Policy, Workflow, and Mission engines for decisions and remediation.

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018 - Operational Lifecycle

Initialize -> Configure -> Validate -> Start -> Process -> Observe -> Degrade Safely -> Stop -> Archive State

Every lifecycle transition shall be logged and testable.

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Quality note: Universal Object Model shall include code-level tests, schema-level tests, and documentation-level traceability. Any deviation discovered during implementation shall be raised as an Engineering Change Request rather than silently changing the architecture.

Operational note: Universal Object Model shall expose health state, readiness state, dependency state, and degraded-mode state. Operators must be able to understand whether the engine is fully operational, partially degraded, blocked by an upstream dependency, or intentionally disabled by policy.

Developer note: implementation of Universal Object Model shall follow clean boundaries between domain logic, infrastructure adapters, API handlers, persistence adapters, and event adapters. This is required so Codex, Claude Code, and human developers can implement and test the module incrementally without hidden coupling.

User-output note: any user-visible output related to Universal Object Model shall be written in clear language first and detailed technical language second. The system must always answer: what was found, why it matters, how AQELYN knows, and what the user can do next.

Government and enterprise note: Universal Object Model shall support auditability, exportable records, retention policy alignment, and review-ready evidence packages. The design supports private users, companies, managed service providers, and government environments without creating separate architectures.

019 - Configuration Model

Configuration Model for Universal Object Model shall be implemented in accordance with AQELYN Platform Architecture Baseline v1.0. The implementation must preserve traceability, deterministic behavior, explainability, and evidence-backed outputs.

This section defines engineering expectations for universal object model and provides the implementation team with stable guidance that should not be bypassed by local optimization or AI-generated redesign.

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020 - Error Handling

Error Handling for Universal Object Model shall be implemented in accordance with AQELYN Platform Architecture Baseline v1.0. The implementation must preserve traceability, deterministic behavior, explainability, and evidence-backed outputs.

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021 - Observability

Observability for Universal Object Model shall be implemented in accordance with AQELYN Platform Architecture Baseline v1.0. The implementation must preserve traceability, deterministic behavior, explainability, and evidence-backed outputs.

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022 - Logging

Logging for Universal Object Model shall be implemented in accordance with AQELYN Platform Architecture Baseline v1.0. The implementation must preserve traceability, deterministic behavior, explainability, and evidence-backed outputs.

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023 - Metrics

Metrics for Universal Object Model shall be implemented in accordance with AQELYN Platform Architecture Baseline v1.0. The implementation must preserve traceability, deterministic behavior, explainability, and evidence-backed outputs.

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024 - Audit and Evidence

Audit and Evidence for Universal Object Model shall be implemented in accordance with AQELYN Platform Architecture Baseline v1.0. The implementation must preserve traceability, deterministic behavior, explainability, and evidence-backed outputs.

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025 - Trust and Confidence

Trust and Confidence for Universal Object Model shall be implemented in accordance with AQELYN Platform Architecture Baseline v1.0. The implementation must preserve traceability, deterministic behavior, explainability, and evidence-backed outputs.

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026 - Policy and Governance

Policy and Governance for Universal Object Model shall be implemented in accordance with AQELYN Platform Architecture Baseline v1.0. The implementation must preserve traceability, deterministic behavior, explainability, and evidence-backed outputs.

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027 - Performance and Scalability

Performance and Scalability for Universal Object Model shall be implemented in accordance with AQELYN Platform Architecture Baseline v1.0. The implementation must preserve traceability, deterministic behavior, explainability, and evidence-backed outputs.

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Government and enterprise note: Universal Object Model shall support auditability, exportable records, retention policy alignment, and review-ready evidence packages. The design supports private users, companies, managed service providers, and government environments without creating separate architectures.

028 - Reliability and Availability

Reliability and Availability for Universal Object Model shall be implemented in accordance with AQELYN Platform Architecture Baseline v1.0. The implementation must preserve traceability, deterministic behavior, explainability, and evidence-backed outputs.

This section defines engineering expectations for universal object model and provides the implementation team with stable guidance that should not be bypassed by local optimization or AI-generated redesign.

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029 - Threat Model

Threat Model for Universal Object Model shall be implemented in accordance with AQELYN Platform Architecture Baseline v1.0. The implementation must preserve traceability, deterministic behavior, explainability, and evidence-backed outputs.

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030 - Secure Implementation Guidance

Secure Implementation Guidance for Universal Object Model shall be implemented in accordance with AQELYN Platform Architecture Baseline v1.0. The implementation must preserve traceability, deterministic behavior, explainability, and evidence-backed outputs.

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031 - Testing Strategy

Testing shall include unit tests, integration tests, contract tests, security tests, performance tests, regression tests, and acceptance tests. Tests must be runnable locally and in CI/CD.

- Unit tests validate pure logic and boundary conditions.
- Integration tests validate engine interactions.
- Contract tests validate API and event schemas.
- Security tests validate authorization, input handling, and secret safety.

Implementation note: Universal Object Model must treat every decision as auditable. If the engine produces a finding, state transition, recommendation, event, or policy result, that output must reference the reason, source, timestamp, and confidence context. This supports AQELYN's user promise that security outcomes are understandable to non-experts and actionable by experts.

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032 - Validation Strategy

Validation Strategy for Universal Object Model shall be implemented in accordance with AQELYN Platform Architecture Baseline v1.0. The implementation must preserve traceability, deterministic behavior, explainability, and evidence-backed outputs.

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033 - Acceptance Criteria

Acceptance Criteria for Universal Object Model shall be implemented in accordance with AQELYN Platform Architecture Baseline v1.0. The implementation must preserve traceability, deterministic behavior, explainability, and evidence-backed outputs.

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034 - Requirements Matrix

ID	Requirement	Priority	Verification
---	---	---	---
FR-001	Capability 1 for Universal Object Model	Mandatory	Test + Review
FR-002	Capability 2 for Universal Object Model	Mandatory	Test + Review
FR-003	Capability 3 for Universal Object Model	Mandatory	Test + Review
FR-004	Capability 4 for Universal Object Model	Mandatory	Test + Review
FR-005	Capability 5 for Universal Object Model	Mandatory	Test + Review
FR-006	Capability 6 for Universal Object Model	Mandatory	Test + Review
FR-007	Capability 7 for Universal Object Model	Mandatory	Test + Review
FR-008	Capability 8 for Universal Object Model	Mandatory	Test + Review
FR-009	Capability 9 for Universal Object Model	Mandatory	Test + Review
FR-010	Capability 10 for Universal Object Model	Mandatory	Test + Review

035 - Traceability Matrix

Requirement	Component	Test	Evidence
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FR-001	Universal Object Model Service Layer	automated test	evidence reference + CI artifact
FR-002	Universal Object Model Service Layer	automated test	evidence reference + CI artifact
FR-003	Universal Object Model Service Layer	automated test	evidence reference + CI artifact
FR-004	Universal Object Model Service Layer	automated test	evidence reference + CI artifact
FR-005	Universal Object Model Service Layer	automated test	evidence reference + CI artifact
FR-006	Universal Object Model Service Layer	automated test	evidence reference + CI artifact
FR-007	Universal Object Model Service Layer	automated test	evidence reference + CI artifact
FR-008	Universal Object Model Service Layer	automated test	evidence reference + CI artifact
FR-009	Universal Object Model Service Layer	automated test	evidence reference + CI artifact
FR-010	Universal Object Model Service Layer	automated test	evidence reference + CI artifact

036 - Example Artifacts

Example artifacts included with this archive demonstrate canonical payloads, event envelopes, review checklists, and implementation-ready acceptance examples. They are illustrative and may be expanded during coding without changing the architecture.

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037 - Engineering Journal

This archive was regenerated to align early AQELYN foundation packages with the later FULL_COMPLETE publication standard. The regeneration removes legacy AQELYN codename remnants, expands the implementation guidance, and brings PDF/HTML artifacts into consistent layout with EA-0010 through EA-0063.

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038 - Deployment Considerations

Deployment Considerations for Universal Object Model shall be implemented in accordance with AQELYN Platform Architecture Baseline v1.0. The implementation must preserve traceability, deterministic behavior, explainability, and evidence-backed outputs.

This section defines engineering expectations for universal object model and provides the implementation team with stable guidance that should not be bypassed by local optimization or AI-generated redesign.

Implementation note: Universal Object Model must treat every decision as auditable. If the engine produces a finding, state transition, recommendation, event, or policy result, that output must reference the reason, source, timestamp, and confidence context. This supports AQELYN's user promise that security outcomes are understandable to non-experts and actionable by experts.

Quality note: Universal Object Model shall include code-level tests, schema-level tests, and documentation-level traceability. Any deviation discovered during implementation shall be raised as an Engineering Change Request rather than silently changing the architecture.

Operational note: Universal Object Model shall expose health state, readiness state, dependency state, and degraded-mode state. Operators must be able to understand whether the engine is fully operational, partially degraded, blocked by an upstream dependency, or intentionally disabled by policy.

Developer note: implementation of Universal Object Model shall follow clean boundaries between domain logic, infrastructure adapters, API handlers, persistence adapters, and event adapters. This is required so Codex, Claude Code, and human developers can implement and test the module incrementally without hidden coupling.

User-output note: any user-visible output related to Universal Object Model shall be written in clear language first and detailed technical language second. The system must always answer: what was found, why it matters, how AQELYN knows, and what the user can do next.

Government and enterprise note: Universal Object Model shall support auditability, exportable records, retention policy alignment, and review-ready evidence packages. The design supports private users, companies, managed service providers, and government environments without creating separate architectures.

039 - Codex and AI Implementation Guidance

Codex and AI Implementation Guidance for Universal Object Model shall be implemented in accordance with AQELYN Platform Architecture Baseline v1.0. The implementation must preserve traceability, deterministic behavior, explainability, and evidence-backed outputs.

This section defines engineering expectations for universal object model and provides the implementation team with stable guidance that should not be bypassed by local optimization or AI-generated redesign.

Implementation note: Universal Object Model must treat every decision as auditable. If the engine produces a finding, state transition, recommendation, event, or policy result, that output must reference the reason, source, timestamp, and confidence context. This supports AQELYN's user promise that security outcomes are understandable to non-experts and actionable by experts.

Quality note: Universal Object Model shall include code-level tests, schema-level tests, and documentation-level traceability. Any deviation discovered during implementation shall be raised as an Engineering Change Request rather than silently changing the architecture.

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Government and enterprise note: Universal Object Model shall support auditability, exportable records, retention policy alignment, and review-ready evidence packages. The design supports private users, companies, managed service providers, and government environments without creating separate architectures.

040 - Publication Readiness

EA-0002 is publication-ready when the master Markdown, PDF, HTML, README, manifest, requirements matrix, traceability matrix, engineering journal, examples, diagrams, and release ZIP are present and hash-verified.

Implementation note: Universal Object Model must treat every decision as auditable. If the engine produces a finding, state transition, recommendation, event, or policy result, that output must reference the reason, source, timestamp, and confidence context. This supports AQELYN's user promise that security outcomes are understandable to non-experts and actionable by experts.

Quality note: Universal Object Model shall include code-level tests, schema-level tests, and documentation-level traceability. Any deviation discovered during implementation shall be raised as an Engineering Change Request rather than silently changing the architecture.

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